

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME			
CLASS		INDEX NUMBER	

BIOLOGY

9648/02

Paper 2 Core Paper

29 August 2017

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the question paper.

Section B

Answer any **one** question on the answer paper provided.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	11
2	10
3	12
4	12
5	12
6	11
7	12
Section B	
8 OR 9	20
Total	100

This document consists of **19** printed pages and **1** blank page.



Section A
Answer **all** questions.

- 1 Fig. 1.1 shows the mode of action of sodium-potassium pump found on the cell surface membranes of sweat glands.

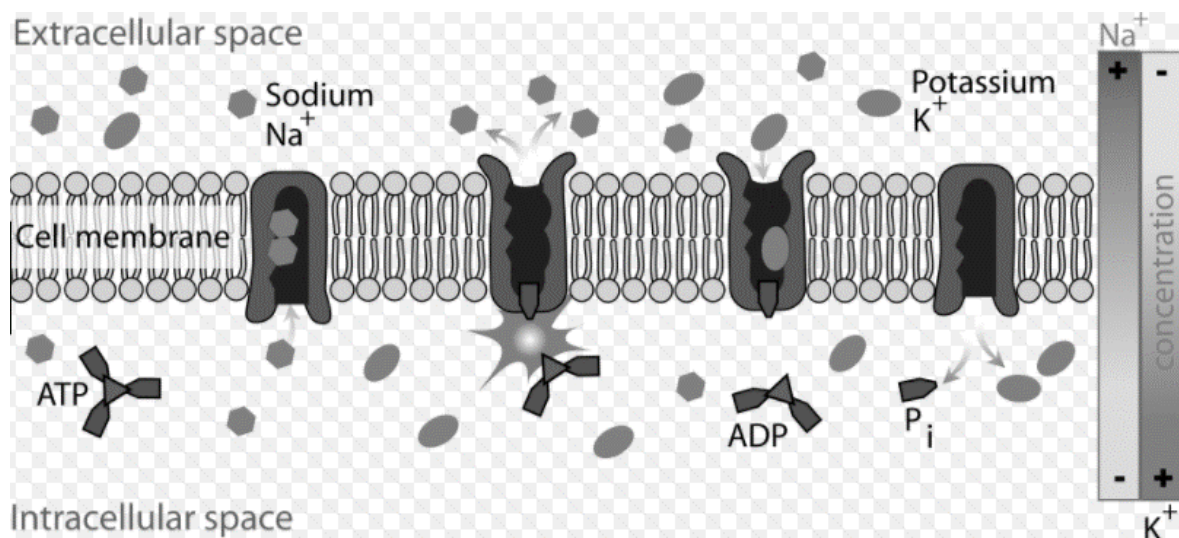


Fig. 1.1

- (a) Explain why the cell surface membrane is referred to as the *fluid mosaic model*.

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..... [2]

- (b) Explain why ions can only move across cell surface membranes via transmembrane proteins.

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..... [2]

The sodium-potassium pump, also known as sodium-potassium ATPase, is able to catalyse the hydrolysis of ATP molecules to ADP and P_i. This reaction is coupled to the movement of ions across the membranes.

Movement of ions is based on the initial binding of Na⁺ ions to the pump. This stimulates phosphorylation of the pump which causes a change in its conformation, leading to ion movement.

(c) With reference to Fig. 1.1,

- (i) identify the type of transport mechanism that enables sodium ions to leave the cell;

..... [1]

- (ii) explain how the pump is also able to catalyse the hydrolysis of ATP molecules leading to the movement of ions across the membranes.

..... [3]

Ouabain is a cardiac glycoside that is able to affect the sodium-potassium pump. In an experiment to determine the effect of ouabain on the pump, sweat glands from rhesus monkeys were taken and treated to measure ATP hydrolysis.

For each of the treatments, 5 sweat gland units were placed in a mixture of buffer and ATP.

Treatment 1: Sweat glands of rhesus monkey with Na^+ and K^+ added.

Treatment 2: Sweat glands of rhesus monkey with Na^+ , K^+ and ouabain added.

Treatment 3: Sweat glands of rhesus monkey without additional Na^+ and K^+ .

Table 1 shows the data collected at the end of the 1 hour experiment.

Table 1

Treatment	Number of moles of ATP hydrolysed/ g^{-1}
1	0.58
2	0.37
3	0.38

- (d) Explain the effects of ouabain on ATP hydrolysis.

..... [2]

- (e) Suggest a reason for the results of Treatment 3.

..... [1]

[Total: 11]

- 2 Fig 2.1 is a photomicrograph of plants cells, with some undergoing mitosis.

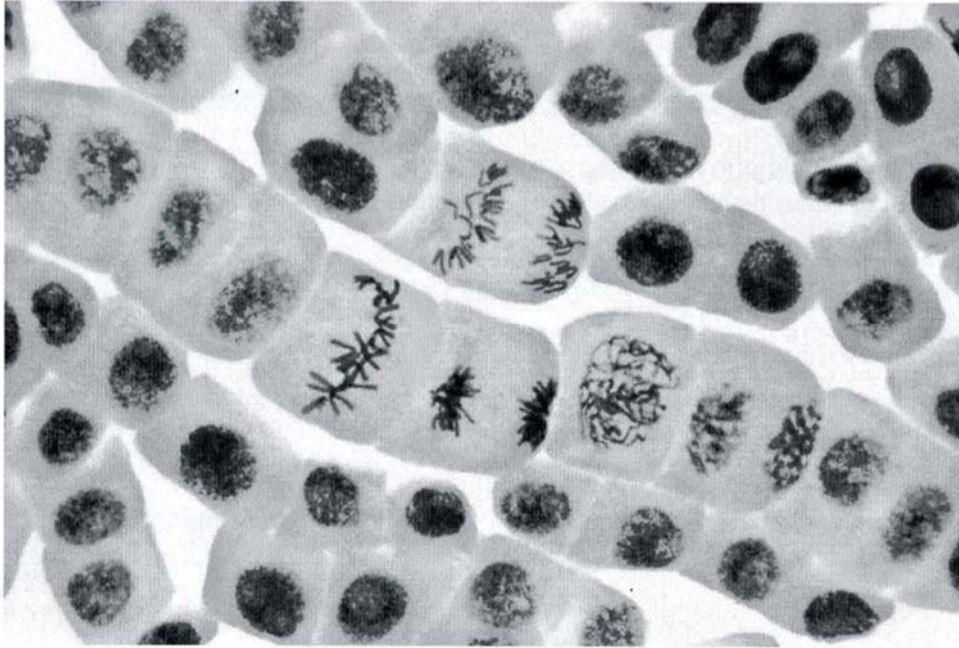


Fig. 2.1

- (a) On Fig 2.1, use labels and label lines to indicate **one** cell in anaphase stage of mitosis. [1]
- (b) The longest stage of the mitotic cell cycle, interphase, is divided into three phases, **G1, S** and **G2**.
- (i) Describe what happens in the G1 phase.

..... [3]

- (ii) There are various checkpoints in the mitotic cell cycle. One of them is present in the G1 phase called the G1 checkpoint.

Describe the function of G1 checkpoint.

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..... [2]

- (iii) When cell cycle checkpoints are defective, cancer could arise.

With reference to two named genes, outline the development of cancer.

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..... [4]

[Total: 10]

- 3 Fig. 3.1 shows a scanning electron micrograph of *Escherichia coli* under mass attack by numerous phage-T4 virions.

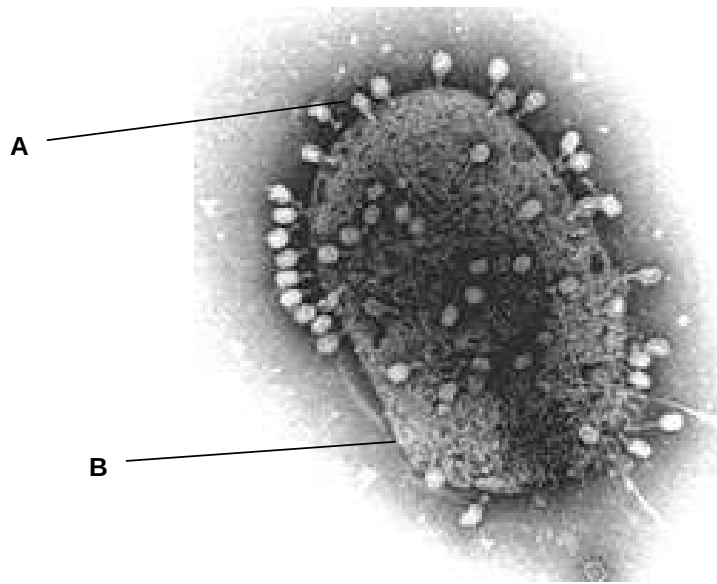


Fig. 3.1

- (a) Identify and briefly describe the function of the structures labelled **A** and **B** in Fig. 3.1.

.....

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.....

..... [2]

- (b) Describe how T4 phage attaches and penetrates into *Escherichia coli*.

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..... [3]

Lambda (λ) phage infections result in the lysis of the bacterial host cells which shows up as a plaque or a zone of clearing on an agar plate containing a dense growth of bacteria.

This observation can be used to quantify the number of infectious bacteriophage particles through a plaque assay. 'Plaque Forming Unit (PFU)' gives the measurement of the number of infectious viral particles present during the experiment.

Fig. 3.2 below shows the changes in PFU caused by lambda (λ) phages infecting *E.coli* cells upon irradiation by UV.

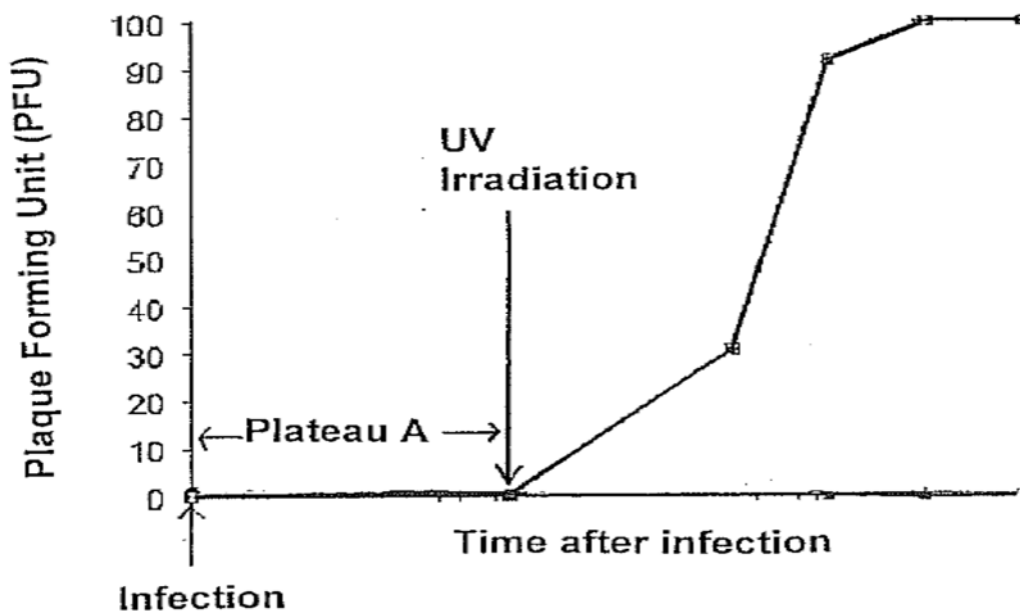


Fig. 3.2

- (c) State the difference in the type of life cycle between T4 and λ phage.

[1]

- (d) With reference to Fig. 3.2,

- (i) suggest an explanation for the flat part of the graph, labelled 'Plateau A'.

[3]

- (ii) Account for the rise in PFU shortly after UV irradiation.

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..... [3]

[Total: 12]

- 4 The flowers of a species of *Primula* may have a yellow 'eye' in the centre of the flower or be 'eyeless'. The 'eye' may be small or large, as shown in Fig. 4.1.

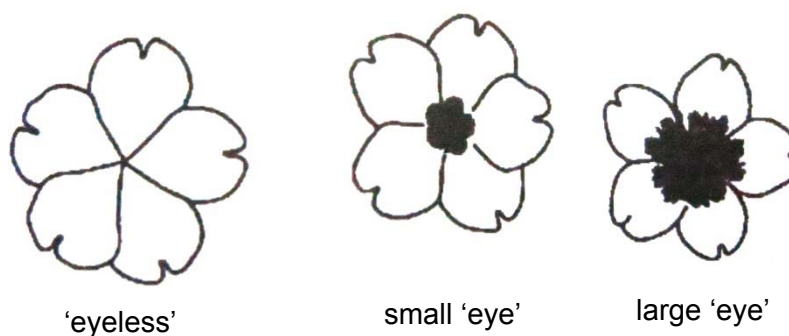


Fig. 4.1

These phenotypes are produced by three alleles of a gene. Allele A^S suppresses yellow pigment production in the 'eye' giving 'eyeless' flowers. Allele A^S is dominant to allele A^R , which restricts the size of the eye, resulting in flowers with small 'eyes'. Both A^S and A^R are dominant to a .

- (a) (i) Explain what is meant by dominant allele.

.....

..... [1]

- (ii) List all the genotypes that give rise to each type of flower.

Phenotypes	Genotypes
'eyeless' flowers	
"small-eyed" flowers	
'large-eyed' flowers	

[3]

- (iii) Suggest a mechanism by which allele A^S suppresses yellow pigment production.

[2]

- (b) A plant with the genotype $A^S a$ was crossed with one with the genotype $A^R a$.

Draw a genetic diagram to show the gametes and the genotypes and phenotypes of the offspring of this cross. Give the expected ratio of offspring phenotypes.

[2]

- (c) In such a cross between plants with genotypes $A^S a$ and $A^R a$, the following numbers of phenotypes of offspring were recorded.

phenotype of flower	number of offspring
'eyeless' flowers	85
'small-eyed' flowers	59
'large-eyed' flowers	56

- (i) Use the chi-squared (χ^2) test and the table of probabilities shown in the **Table 4** to find the probability of the results of this cross departing significantly by chance from expectation. Show your working.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

O = observed result
E = expected result
 $v = n - 1$

Table 4

degrees of freedom	probability P				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.42	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

[2]

- (ii) State the conclusion that could be drawn from the probability found in **c(i)**.

[2]

[Total: 12]

- 5 Fig 5.1 shows a ligand binding to the G Protein-coupled receptor (GPCR) which is embedded on the cell surface membrane.

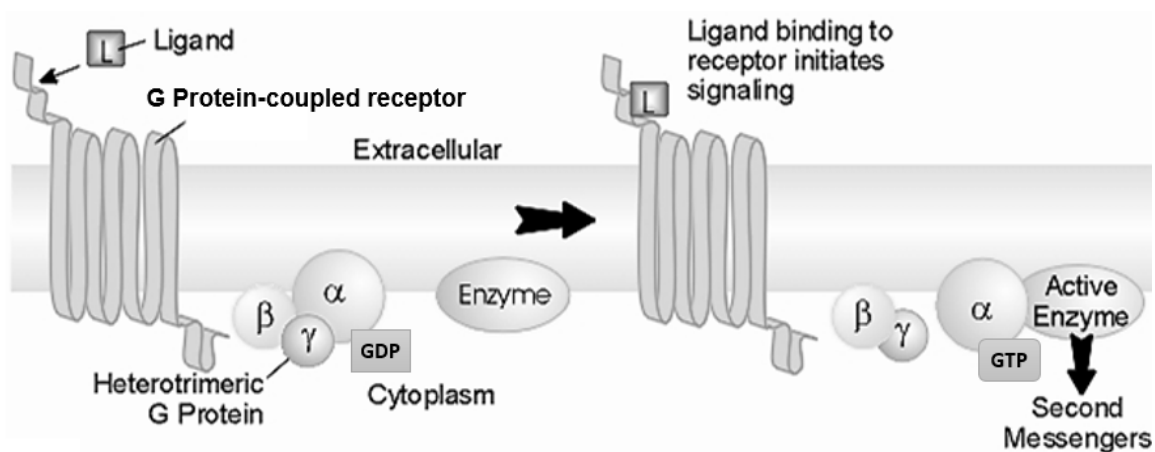


Fig. 5.1

- (a) (i) Identify the ligand in Fig. 5.1.

..... [1]

- (ii) Explain why ligand mentioned in **a(i)** unable to pass through the cell surface membrane.

.....

 [2]

- (b) With reference to Fig 5.1, describe what happens when the ligand binds to the GPCR.

.....

 [3]

- (c) With a named example, define “second messengers”.

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..... [2]

- (d) Explain how intracellular signal is terminated when ligand is released from the receptor.

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..... [2]

- (e) One of the side effects of a particular drug includes non-responsiveness of GPCR to ligands.

Suggest how the drug could have caused such non-responsiveness of GPCR to ligands.

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.....

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..... [2]

[Total: 12]

- 6 ATP synthase is an enzyme that synthesizes ATP from ADP and inorganic phosphate (P_i).

Fig. 6.1 shows the structure of ATP synthase. It is found in both the chloroplast and the mitochondrion.

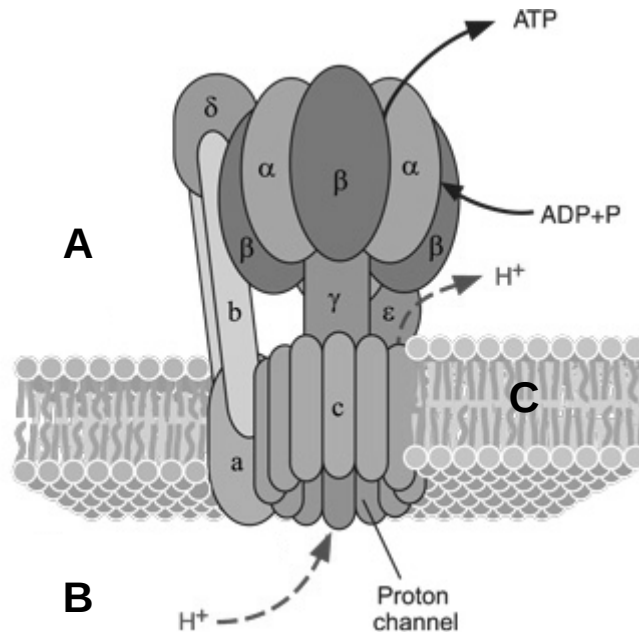


Fig. 6.1

- (a) In Table 6.1, indicate the regions A, B and C that are found in mitochondrion and chloroplast respectively.

Table 6.1

organelle	A	B	C
chloroplast			
mitochondrion			

[2]

In both organelles, a proton gradient exists across the membrane where ATP synthase is located. The synthesis of ATP by ATP synthase is coupled with the diffusion of H^+ across the proton channel down its concentration gradient.

(b) Describe how this proton gradient is generated in the chloroplast.

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..... [3]

The blue dye DCPIP can be converted to colourless reduced DCPIP by gaining electrons.

A suspension of chloroplasts was made by grinding fresh leaves in buffer solution and centrifuging the mixture. Tubes were then prepared and treated as shown in Table 6.2.

Table 6.2

Tubes	Content	Treatment	Colour	
			Start	After 20 mins
A	1 cm ³ chloroplast suspension 5cm ³ DCPIP	Illuminated strongly	Blue-green	Green
B	1 cm ³ buffer solution 5cm ³ DCPIP	Illuminated strongly	Blue	Blue
C	1 cm ³ chloroplast suspension 5cm ³ DCPIP	Left in the dark	Blue-green	Blue-green

(c) With reference to Table 6.2,

(i) name a possible source of electrons that reduces DCPIP;

..... [1]

- (ii) explain this observation using your knowledge on light-dependent reactions of photosynthesis.

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..... [3]

- (d) Paraquat is a weedkiller. It binds with electrons of PS I.

Suggest how paraquat results in the death of a plant.

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..... [2]

[Total: 11]

- 7 The marine threespine sticklebacks, *Gasterosteus aculeatus* is a freshwater fish living in the lakes of British Columbia, Canada as shown in Fig. 7.1.

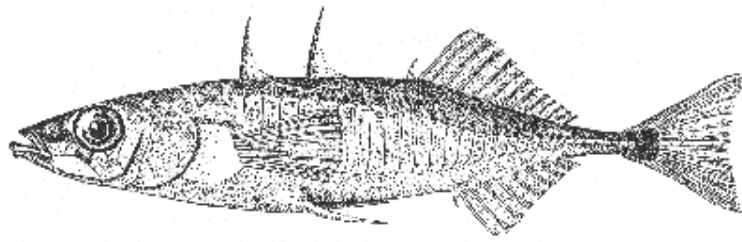


Fig. 7.1

In order to investigate the process of speciation in these populations, three small lakes were studied. Each lake contained two varieties of stickleback: a large, bottom-dwelling variety that fed on invertebrates near the shore and a small, plankton-eating variety that lived in the open water. The probability of breeding between pairs of individuals was measured under laboratory conditions in the following breeding combinations:

I	different varieties from the same lake
II	different varieties from different lakes
III	same variety from different lakes
IV	same variety from the same lake

The data are summarized in Fig. 7.2 below.

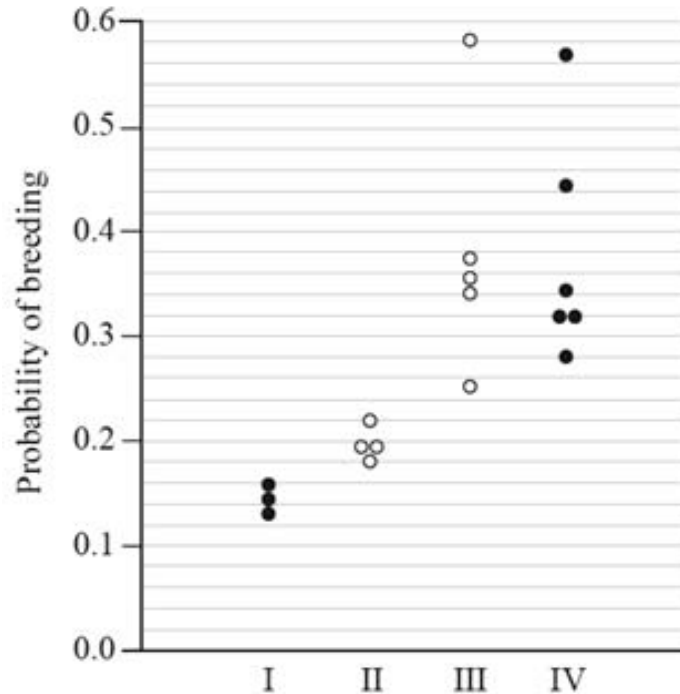


Fig. 7.2

(a) With reference to Fig. 7.2,

- (i)** identify the highest and lowest probabilities of breeding for individuals of the same variety;

..... [1]

- (ii)** describe the differences in probability of breeding between individuals from different lake;

..... [2]

- (iii)** describe the evidence that speciation is taking place in these populations and explain the type of speciation as shown in Fig. 7.2;

..... [3]

- (iv)** explain why all the individuals are still considered the same species.

..... [2]

- (b) The freshwater lakes also contain many different types of parasites that infect the different varieties of marine threespine sticklebacks.

Explain why these parasites enable speciation of the marine threespine sticklebacks to occur.

[4]

[Total: 12]

Section BAnswer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section **(a)**, **(b)** etc., as indicated in the question.

8 (a) Compare prokaryotic and eukaryotic genomes. [8]

(b) Explain how the expression of a gene can be decreased by chromatin modification. [6]

(c) Explain why processing of pre-mRNA is necessary for regulation of gene expression. [6]

[Total: 20]

9 (a) Using *lac* operon as an example, describe how prokaryotes such as *Escherichia coli* is able to respond to changes in the environment. [8]

(b) Contrast binary fission and mitosis. [6]

(c) Explain the significance of genetic exchange in bacteria. [6]

[Total: 20]

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